

Unit 9, Lesson 9: Scale Factors of Dilations

Benchmarks

MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.

MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

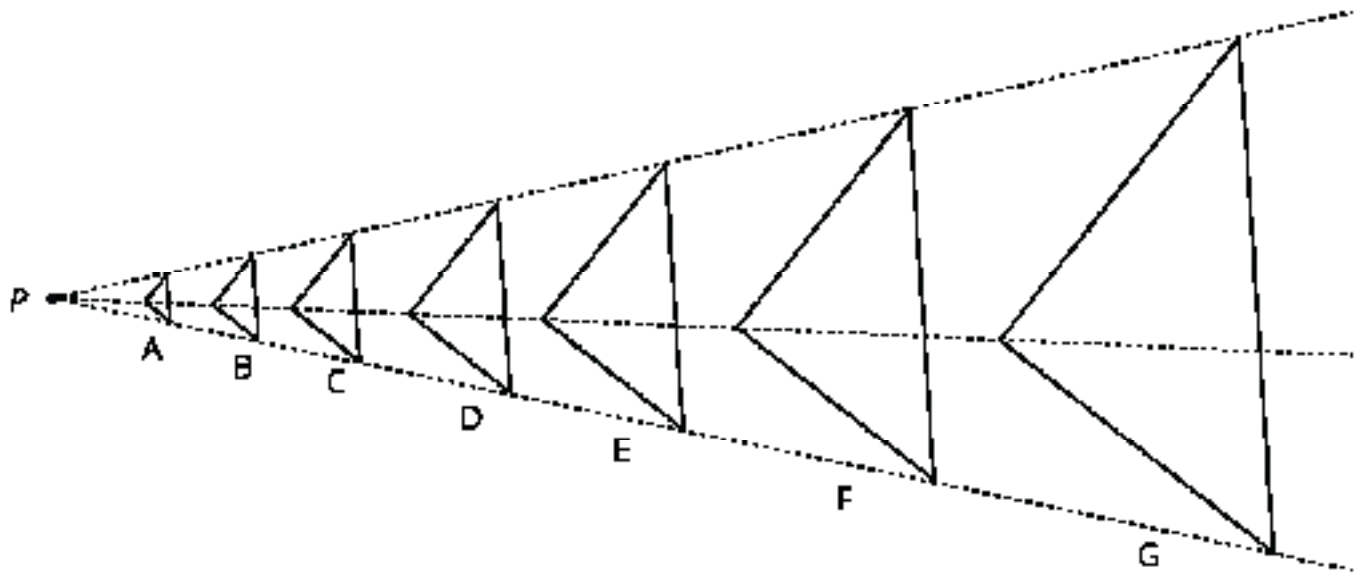
MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

Learning Targets

- ☐ I can determine the scale factor of a given dilation.
- ☐ I can solve real-world problems that involve similar triangles.

9.9.1 Warm-Up: Bell Work

1. All of the triangles shown are dilations of Triangle *D*. The dilations use the same center, point *P*, but different scale factors.

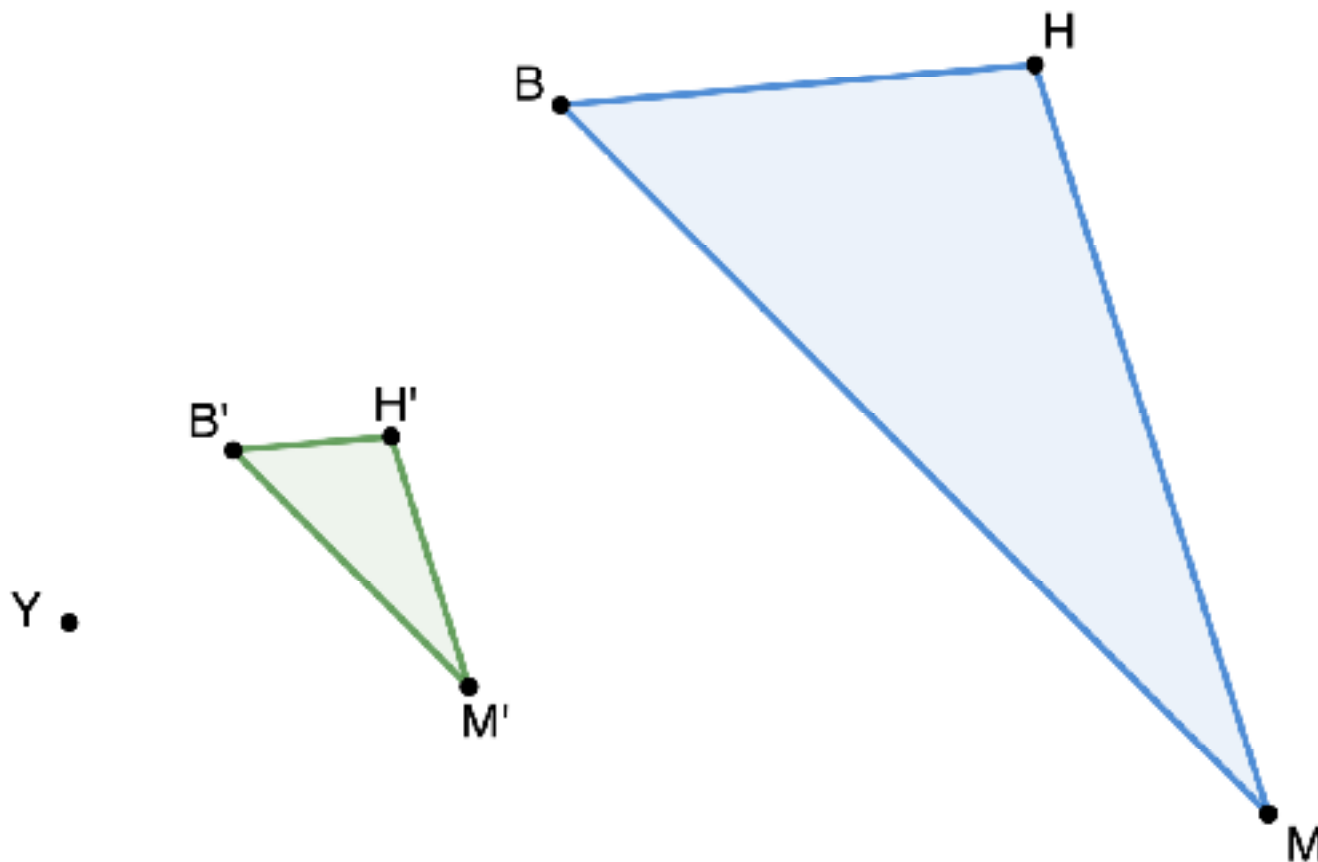


- How do Triangles *A*, *B*, and *C* compare to the preimage?
- How do Triangles *E*, *F*, and *G* compare to the preimage?
- What do you know about the different scale factors used based on your answers to Parts A and B?

9.9.2 Exploration: Identifying the Scale Factor of Dilations

Work with your partner to complete the following.

1. Triangle BHM is dilated using point Y as the center, as shown.



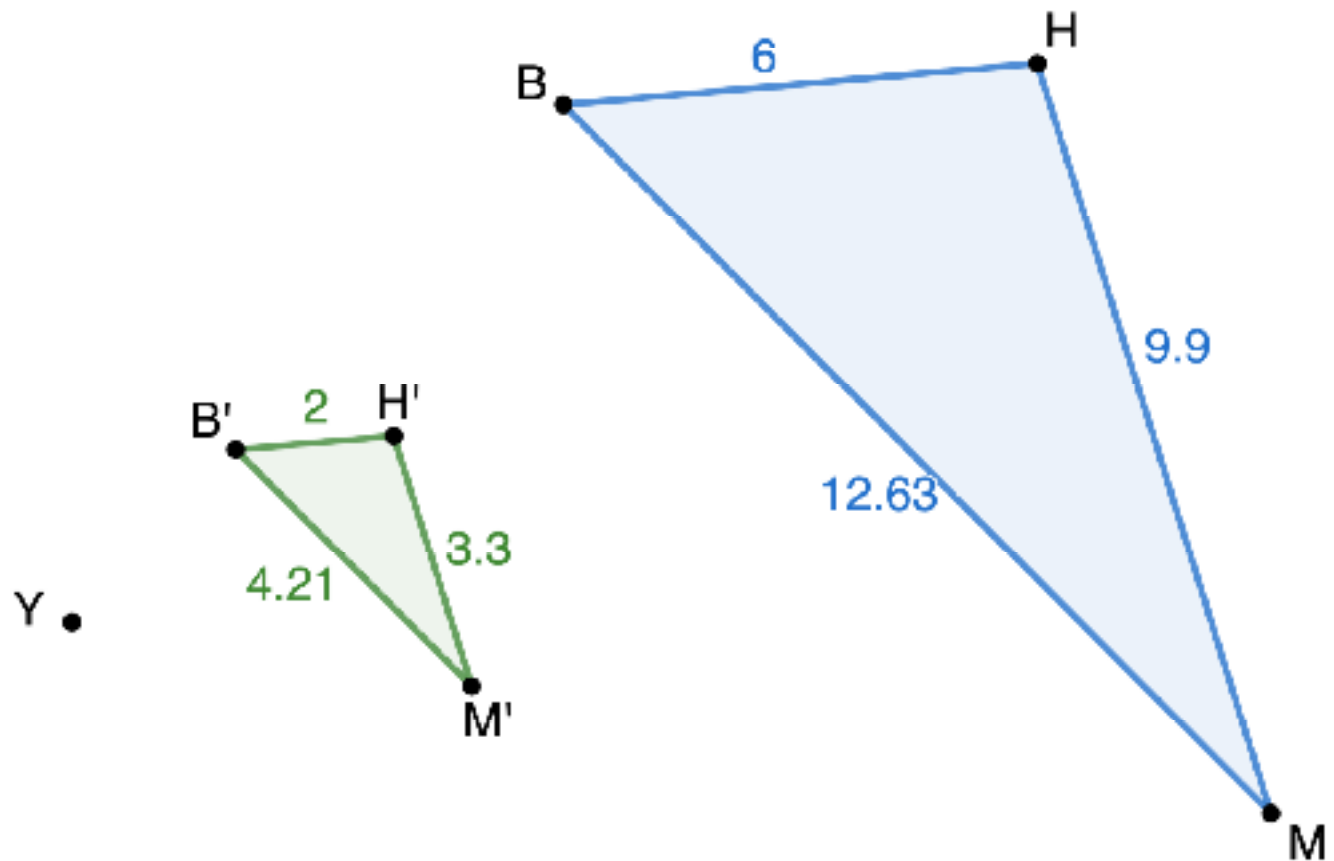
- a. Complete the statement.

The scale factor of this dilation is

- ☐ between 0 and 1.
- ☐ equal to 1.
- ☐ greater than 1.

- b. Explain your reasoning.

- c. The measurements, in units, of the side lengths of each triangle have been added to the figure.

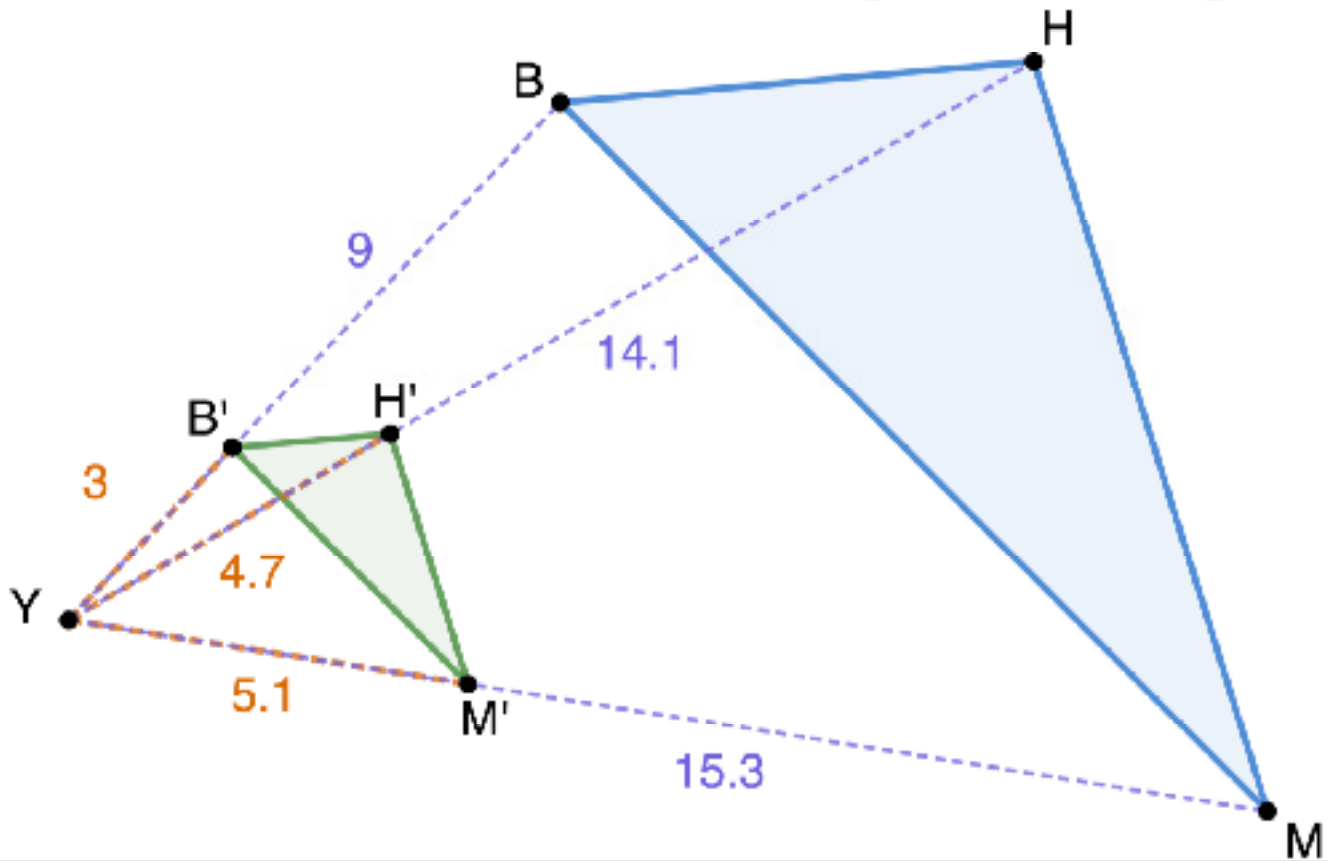


Myrical said the scale factor used is $\frac{1}{3}$, while Joshua said it is 3. Explain who is correct.

- d. Discuss with your partner how the distance between points Y and M should compare to the distance between points Y and M' . Then, complete the statement.

The distance between points Y and M should be ____ times the distance between Y and M' .

- e. The distances, in units, between the center of dilation and each vertex on the preimage are shown in purple. The distances, in units, between the center of dilation and each vertex on the image are shown in orange.



Explain whether the information given confirms your answer to Part D.

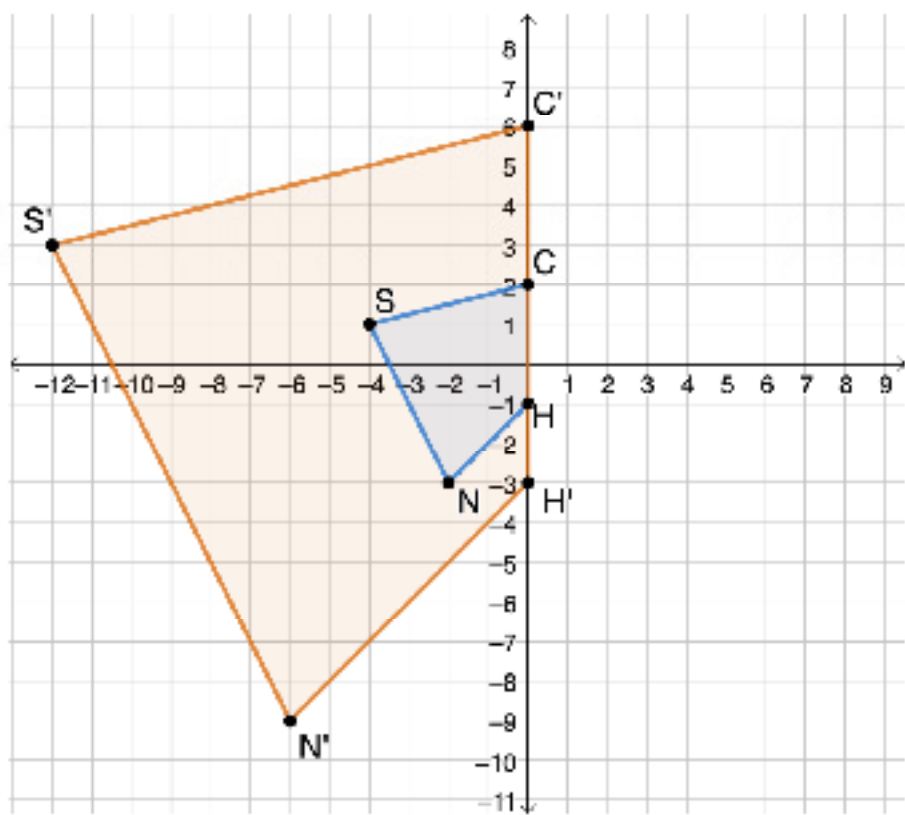


The scale factor of a dilation can be determined using the lengths of the corresponding sides of the preimage or using the distances between the center of dilation and the vertices of the preimage and image.

$$\text{Scale factor} = \frac{\text{corresponding side length of image}}{\text{corresponding side length of preimage}}$$

$$\text{Scale factor} = \frac{\text{distance from center to vertex on the image}}{\text{distance from center to corresponding preimage vertex}}$$

2. Quadrilateral *CHNS* is dilated using the origin as the center, as shown.



a. Complete the table.

Vertices of the Preimage	Coordinates	Vertices of the Image	Coordinates
Point <i>C</i>		Point <i>C'</i>	
Point <i>H</i>		Point <i>H'</i>	
Point <i>N</i>		Point <i>N'</i>	
Point <i>S</i>		Point <i>S'</i>	

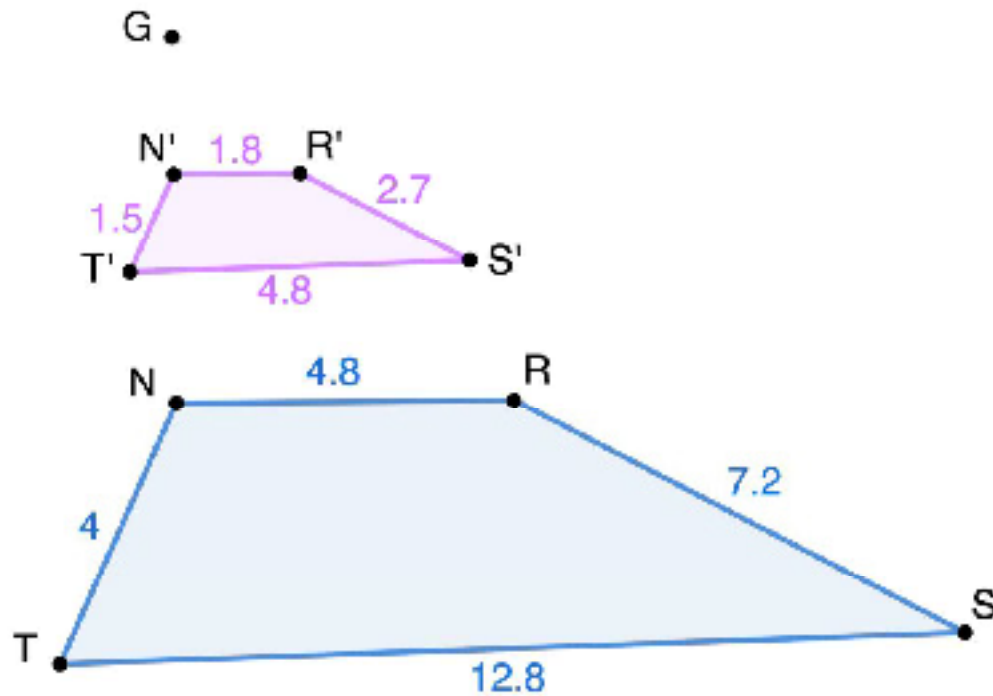
b. Describe the relationship between the coordinates of the vertices on the preimage and the corresponding coordinates on the image.

c. What is the scale factor of the dilation?

9.9.3 Collaboration: Error Analysis

Work with your partner to complete the following.

1. Quadrilateral $NRS'T$ is dilated using point G as the center of dilation.

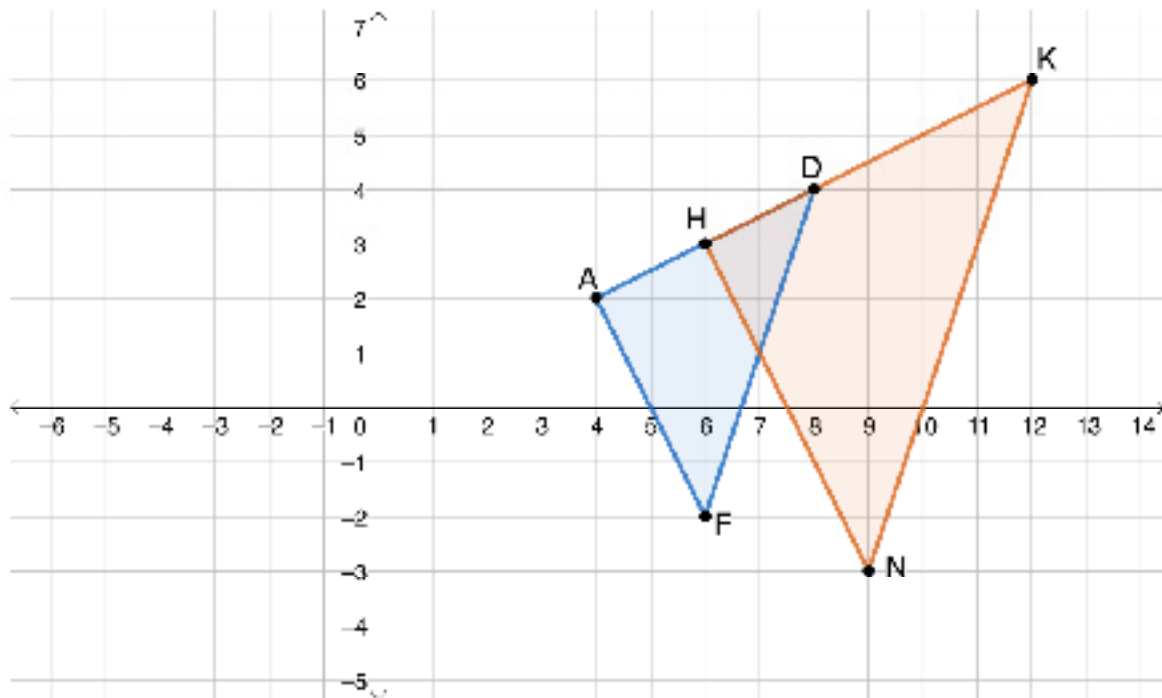


Latonya found the scale factor of dilation to be $\frac{8}{3}$. Her work is shown.

$$\text{Scale Factor} = \frac{12.8}{4.8} = \frac{8}{3}$$

- a. Is Latonya correct?
- b. Explain your reasoning.

2. On the coordinate grid, $\triangle HKN$ is the result of dilating $\triangle ADF$ using the origin as the center of dilation.

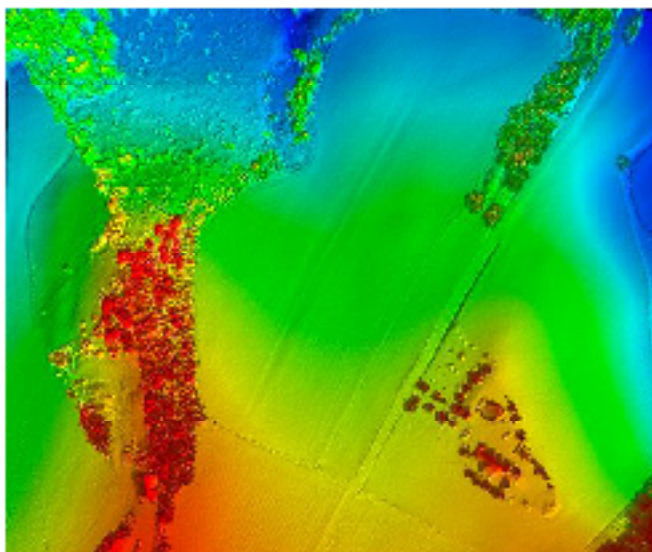


Martell says, "The scale factor of this dilation is 3. To find the scale factor, I found the coordinates of point A which were (4, 2), and then I found the coordinates of point K which were (12, 6). I know the scale factor is 3 because $4 \times 3 = 12$ and $2 \times 3 = 6$."

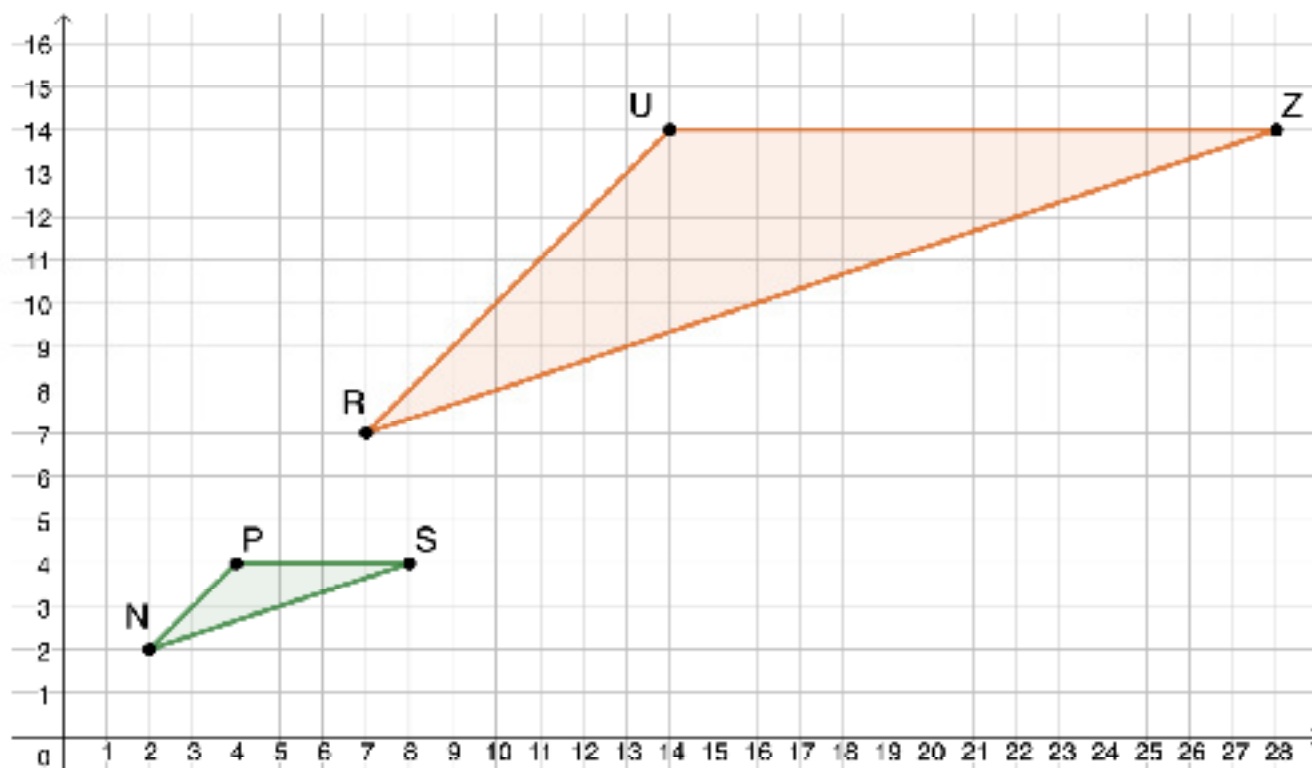
- a. Explain the error Martell made.
- b. What scale factor was used to dilate $\triangle ADF$?

9.9.4 Guided Practice: Dilations in the Real World

1. Archaeologists have begun using LiDAR (light detection and ranging) technology to map the shape of the ground underneath dense rainforests and jungles. Using this technology, they have discovered numerous ancient cities that have been hidden for thousands of years.



An archaeologist discovered two triangular structures using LiDAR. She overlaid her image on a coordinate grid to analyze the relationship between the structures.

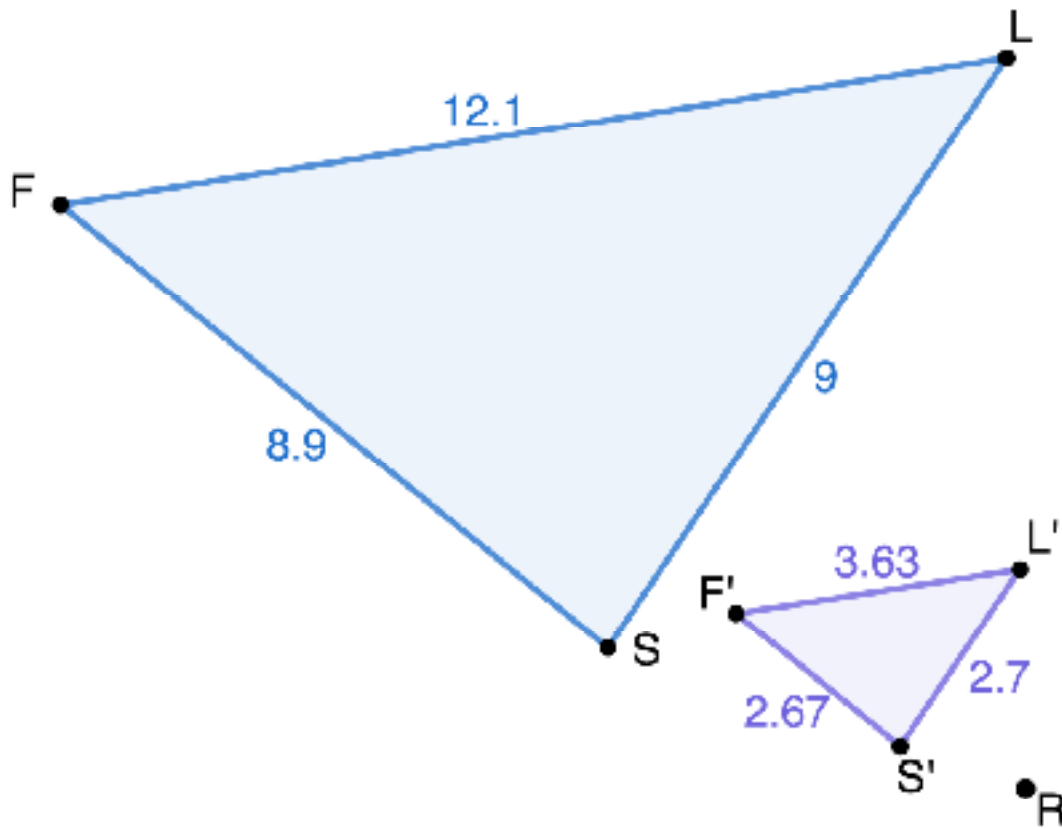


- a. The archaeologist found that the larger triangle, $\triangle RYZ$, could be obtained by dilating the smaller triangle, $\triangle NPS$, using the origin as the center of dilation. Identify the scale factor of the dilation.
- b. Suppose the archaeologist found a third structure that is a dilation of $\triangle NPS$ using a scale factor of 11.5 centered at the origin. Determine what the coordinates of the third structure, $\triangle N'P'S'$, would be.

Vertex	Coordinates
N'	
P'	
S'	

9.9.5 Wrap-Up: Ticket Out the Door

1. The figure shows the result of dilating $\triangle FLS$ using point R as the center of dilation. Measurements, in units, of each side are given.



Determine the scale factor of the dilation. Show your work.

